

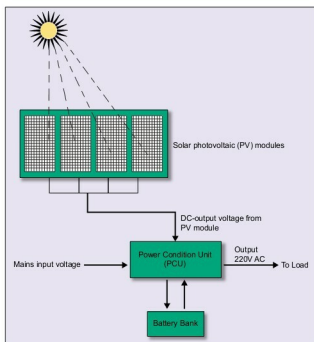


Fig. 4: Solar-home system

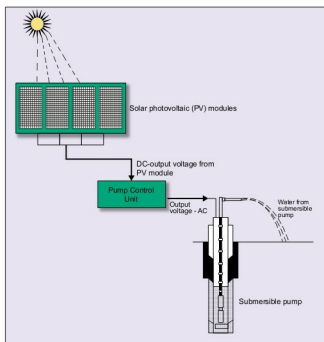


Fig. 5: Solar water pump

circuit in the PCU. The charge control circuit controls the voltage and current and sends these to the battery bank to charge the battery. The inverter in the PCU processes the 12-24V DC voltage into the required output voltage of 220V AC at 50Hz, which is connected to the load.

The system has a microcontroller-based pulse width modulation (PWM) charge controller which can control the charging current. When the solar energy from the solar panel is available, the mains power will not be used for battery charging or for the load. The load takes power from the solar panel. Power from the mains will be used only when the battery voltage is below the preset voltage (this voltage can be preset). When mains power is on, mains supply and power from the solar unit does not load each other. The charging is carried out with the intelligently-shared current.

An important fact is that if the PCU gets overloaded whilst working on solar, the load gets shifted to mains supply automatically and again when the battery charge reaches the preset level, mains supply is cut off and the load gets shifted to solar/battery again. Battery deep discharge, overcharge

protections are incorporated inside the PCU. This ensures health of battery thereby extending its life.

Solar water-pumping system

Solar water-pumping system is specifically designed to pump water in areas where there is no reliable supply of electricity. The system is used to pump water for domestic use, agricultural irrigation, aquaculture industry, water treatment, etc.

Solar water-pumping system comprises a set of solar PV modules. The PV modules absorb sunlight energy and generate required level of DC voltage and current in proportion to the solar PV modules and available sunlight. The DC output voltage produced by the PV modules is sent to the pump control unit. The pump control unit processes the DC voltage into required levels of three-phase/one-phase AC voltage for the operation of the pumping system. Drive circuits are also integrated in the pump control unit, so no other additional devices are required either to start or run the motor. The system is also equipped with protection circuits to work against low voltage, overvoltage, overcurrent,

overheat, etc.

In the pump control unit, the most important part is solar-pump inverter which has the maximum power point tracking (MPPT) function. This inverter adjusts the output frequency automatically to achieve MPPT according to the intensity of the sunlight.

With solar water-pumping system, the electricity bills and operating and maintenance costs can be minimised or eliminated.

Solar water-heating system

Solar water-heating systems are now widely used. Two types of solar water heating systems are popular. Evacuated tube collector (ETC) and flat plate collector (FPC). The collector assembly is installed at an angle facing the south direction. A storage tank is installed at a height behind the collector assembly and it is connected with a connecting pipe to the collector assembly. The water heating system is installed outdoors in an open space and it is connected to a continuous water supply from the overhead tank. Water flows in the collector assembly through tubes, absorbs heat from the solar panel and hot water is produced. This water is stored in the storage tank installed behind the col-